

just for that, called Omega TradeStation™ that is currently on the market, in use by tens of thousands of traders. It even offers its own computer language. Beset with insomnia, the computerized day-traders become night-testers plowing the data for some of its properties. By dint of throwing their monkeys on typewriters, without specifying what book they want their monkey to write, they will hit upon hypothetical gold somewhere. Many of them blindly believe in it.

One of my colleagues, a man with prestigious degrees, grew to believe in such a virtual world to the point of losing all sense of reality. Whether the modicum of common sense left in him might have rapidly vanished under the mounds of simulations, or whether he might have had none to engage in such pursuit, I cannot tell. By closely watching him I learned that what natural skepticism he may have had vanished under the weight of data – for he was extremely skeptical, but in the wrong area. Ah, Hume!

A MORE UNSETTLING EXTENSION

Historically, medicine has operated by trial and error – in other words, statistically. We know by now that there can be entirely fortuitous connections between symptoms and treatment, and that some medications succeed in medical trials for mere random reasons. I cannot claim expertise in medicine, but have been a steady reader of a segment of the medical literature over the past half decade, long enough to be concerned with the standards, as we will see in the next chapter. Medical researchers are rarely statisticians; statisticians are rarely medical researchers. Many medical researchers are not even remotely aware of this bias. True, it may play a small role, but it is certainly present. One recent medical study links cigarette smoking to a *reduction* in breast cancer, thus conflicting with all previous studies. Logic would indicate that the result may be suspicious, the result of mere coincidence.

THE EARNINGS SEASON: FOOLED BY THE RESULTS

Wall Street analysts, in general, are trained to find the accounting tricks that companies use to hide their earnings. They tend to beat the companies

at that game. But they are not trained yet to deal with randomness. When a company shows an increase in earnings once, it draws no immediate attention. Twice, and the name starts showing up on computer screens. Three times, and the company will merit some buy recommendation.

Just as with the track record problem, consider a cohort of 10,000 companies that are assumed on average to barely return the risk-free rate (i.e. Treasury bonds). They engage in all forms of volatile businesses. At the end of the first year, we will have 5,000 “star” companies showing an increase in profits (assuming no inflation), and 5,000 “dogs”. After three years, we will have 1,250 “stars”. The stock review committee at the investment house will give your broker their name as a “strong buy”. He will leave a voice-message that he has a hot recommendation that necessitates immediate action. You will be e-mailed a long list of names. You will buy one or two of them. Meanwhile, the manager in charge of your 401K retirement plan will be acquiring the entire list.

We can apply the reasoning to the selection of investment categories – as if they were the managers in the example above. Assume you are standing in 1900 with hundreds of investments to look at. There are the stock markets of Argentina, Imperial Russia, the United Kingdom, Unified Germany, and plenty of others to consider. A rational person would have bought not just the emerging country of the United States, but those of Russia and Argentina as well. The rest of the story is well known; while many of the stock markets like those of the United Kingdom and the United States fared extremely well, the investor in Imperial Russia would have no better than medium-quality wallpaper in his hands. The countries that fared well are not a large segment of the initial cohort; randomness would be expected to allow a few investment classes to fare extremely well. I wonder if those “experts” who make foolish (and self-serving) statements like “markets will always go up in any 20-year period” are aware of this problem.

CANCER CURES

When I return home from an Asian or European trip, my jet lag often causes me to rise at a very early hour. Occasionally, though very rarely, I